LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO5

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 5

Unit Title: Association Rule Mining and Anomaly Detection

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the part of an association rule that appears before the arrow.

A) The antecedent <-- correct

B) The consequent

C) The support

D) The lift

Answer: A. A rule is written antecedent implies consequent. The antecedent is the set of items already in the basket, the consequent the item predicted to join it.

Q2. Name the structure that FP-Growth builds in order to hold the transaction database in compressed form.

A) The proximity matrix

B) The FP-tree <-- correct

C) The isolation tree

D) The dendrogram

Answer: B. The FP-tree stores shared prefixes of transactions once rather than repeatedly, which is what removes the need for repeated database scans.

Q3. State what the contamination setting of an Isolation Forest specifies.

A) The number of trees that will be grown

B) The depth to which each tree is grown

C) The expected proportion of anomalies in the data <-- correct

D) The number of features used per split

Answer: C. Contamination fixes where the threshold on the anomaly score is placed, and therefore how many records end up flagged.

Q4. Name the anomaly score that compares the density around a record with the density around its neighbours.

A) The silhouette coefficient

B) The Davies-Bouldin Index

C) The confidence of a rule

D) The Local Outlier Factor <-- correct

Answer: D. LOF is a ratio of local densities, which is what allows it to find an outlier inside a sparse region that a global threshold would miss.

Q5. Identify the value of lift at which two items occur together exactly as often as chance would predict.

A) 0

B) 1 <-- correct

C) 0.5

D) Infinity

Answer: B. Lift is the observed co-occurrence divided by the co-occurrence expected under independence, so a value of 1 means the rule adds nothing.

Q6. State the number of passes over the transaction database that FP-Growth requires.

A) One
B) One for each item
C) One for each itemset size
D) Two <-- correct
Answer: D. The first pass counts the individual item frequencies and the second builds the FP-tree; everything after that works on the tree.

Q7. Explain why an Isolation Forest separates an anomalous record after very few random splits.

A) It lies in a sparse region and is easily cut off <-- correct
B) It carries a label marking it as anomalous
C) It is always placed at the root of every tree
D) It has the largest value of every feature
Answer: A. Random splits carve up the space; a point sitting on its own needs few cuts to be isolated, while a point inside a dense cluster needs many.

Q8. Distinguish an itemset from an association rule.

A) An itemset is directed, a rule is an unordered set
B) Both of the two are unordered sets of items
C) An itemset is a set, a rule is a directed implication <-- correct
D) Both of the two are directed implications
Answer: C. Frequent itemsets are found first; rules are then formed by splitting an itemset into an antecedent and a consequent, and a direction matters for confidence.

Q9. Describe what the conviction of a rule measures.

A) How many transactions contain the antecedent
B) How many items appear in the consequent
C) How often the rule would be wrong if the items were independent <-- correct
D) How long the algorithm took to find the rule
Answer: C. Conviction compares the expected frequency of the rule failing under independence with its observed failure rate, so a large value indicates a strong rule.

Q10. Interpret a lift of 0.6 reported for an association rule.

A) The two items occur together less often than by chance <-- correct
B) The two items occur together more often than chance
C) The rule holds in 60 per cent of all transactions
D) The rule was found in 60 per cent of the passes
Answer: A. A lift below one indicates a negative association: buying the antecedent makes the consequent less likely, not more.

Q11. Predict the effect on an Apriori run of lowering the minimum support threshold.

A) Far fewer frequent itemsets and a faster run
B) The same itemsets, found in fewer passes
C) No frequent itemsets can then be found
D) Far more frequent itemsets and a much slower run <-- correct
Answer: D. A lower bar lets many more itemsets survive each pass, and the candidate generation for the next level then grows accordingly.

Q12. Recognize the application described: a system monitors network traffic and raises an alert on a pattern of connections unlike anything it has recorded before.

A) Association rule mining for cross-selling
B) Anomaly detection in cybersecurity <-- correct
C) Market basket analysis of purchases
D) Hierarchical clustering of the traffic
Answer: B. The task is to identify the small number of records unlike the normal bulk, which is exactly anomaly detection applied to intrusion monitoring.

Q13. Explain why the Local Outlier Factor can find an outlier that a single global distance threshold misses.

A) It uses the class labels of the records
B) It examines every pair of records in the table
C) It requires the number of outliers in advance
D) It judges a record against its own neighbourhood's density <-- correct
Answer: D. In a dataset with a dense region and a sparse one, a record can be a clear outlier within the dense region while still lying closer than typical sparse-region spacings.

Q14. Describe what novelty detection assumes about the data it is trained on.

A) That it is fully labelled by an expert
B) That it contains at least half anomalies
C) That it contains no anomalies at all <-- correct
D) That every feature has been standardised
Answer: C. Novelty detection learns what normal looks like from clean data and then flags departures. Anomaly detection instead works on data that already contains the anomalies.

Q15. Explain why the confidence of a rule is misleading when the consequent is a very common item.

A) Confidence cannot be computed for common items
B) A popular item follows almost any antecedent <-- correct
C) Confidence falls as the consequent becomes common
D) Confidence is only defined for rare consequents
Answer: B. If bread appears in most baskets, any rule ending in bread has high confidence without the antecedent telling us anything. Lift is what corrects for this.

Q16. Distinguish the support of a rule from its confidence.

A) Support is a share of all transactions, confidence of the antecedent's <-- correct

B) Confidence is a share of all transactions, support of the antecedent's

C) Both of the two are shares of all the transactions

D) Both of the two are shares of the antecedent's transactions

Answer: A. Support says how often the whole combination appears at all; confidence says how reliable the implication is once the antecedent is present.

Q17. Predict what happens to the number of candidate itemsets as the number of distinct items in the catalogue grows.

A) It grows in direct proportion to the items

B) It grows exponentially with the number of items <-- correct

C) It stays fixed whatever the catalogue holds

D) It falls as more items are added to the catalogue

Answer: B. A catalogue of n items admits 2^n - 1 non-empty itemsets, which is precisely why the anti-monotone pruning rule is essential.

Q18. Interpret a Local Outlier Factor value close to 1 reported for a record.

A) Its density matches that of its neighbours <-- correct

B) It is a clear outlier in the dataset

C) It lies at the exact centre of the data

D) It is a duplicate of another record

Answer: A. LOF near 1 means the record is as densely surrounded as the points around it. Values substantially above 1 are what mark an outlier.

Q19. Solve the following: a database holds 800 transactions and the itemset of milk with bread appears in 120 of them. Determine its support.

A) 0.12

B) 0.20

C) 1.50

D) 0.15 <-- correct

Answer: D. Support is the share of all transactions containing the itemset: 120 / 800 = 0.15.

Q20. Apply the definition of confidence to the rule linking nappies to wipes, given that nappies appear in 400 transactions and nappies with wipes in 260.

A) 0.26

B) 0.40

C) 0.65 <-- correct

D) 1.54

Answer: C. Confidence is the share of antecedent transactions that also hold the consequent: 260 / 400 = 0.65.

Q21. Use the following to determine the lift of a rule: the support of the rule is 0.10, of the antecedent 0.25 and of the consequent 0.40.

A) 1.0 <-- correct

B) 0.4

C) 2.5

D) 0.1

Answer: A. Lift is 0.10 / (0.25 x 0.40) = 0.10 / 0.10 = 1.0, so the two items are independent and the rule adds nothing.

Q22. Compute the number of non-empty itemsets that can be formed from a catalogue of 5 distinct items.

A) 25

B) 32

C) 31 <-- correct

D) 10

Answer: C. The count is 2^5 - 1 = 32 - 1 = 31, since the empty set is excluded.

Q23. Solve the following: determine the number of association rules that can be formed from the frequent itemset holding A, B and C.

A) 3

B) 6 <-- correct

C) 8

D) 9

Answer: B. Each rule splits the three items into a non-empty antecedent and a non-empty consequent, giving 2^3 - 2 = 6 possibilities.

Q24. Apply the Apriori pruning rule to a candidate three-itemset one of whose two-item subsets was found to be infrequent.

A) It is counted in the next database pass

B) It is promoted to the next level directly

C) It is merged with a frequent itemset

D) It is discarded without being counted <-- correct

Answer: D. A superset can never be more frequent than its subsets, so a candidate with an infrequent subset cannot possibly reach the threshold.

Q25. Compute the confidence of a rule for which the support of the combined itemset is 0.12 and the support of the antecedent is 0.30.

A) 0.036

B) 2.5

C) 0.4 <-- correct

D) 0.12

Answer: C. Confidence is the rule support divided by the antecedent support: 0.12 / 0.30 = 0.4.

Q26. Use the following to determine whether items A and B are independent: A has support 0.20, B has support 0.50 and the rule has support 0.10.

A) They are positively associated, the lift exceeding 1
B) They are negatively associated, the lift below 1
C) Independence cannot be judged from these figures
D) They are independent, since the lift is exactly 1 <-- correct

Answer: D. The expected joint support under independence is 0.20 x 0.50 = 0.10, exactly what is observed, so the lift is 1.

Q27. Examine a rule with confidence 0.90 and lift 1.05, and determine what should be concluded.

A) The consequent is very common and the rule adds little <-- correct
B) The rule is among the strongest that can be found
C) The confidence must have been computed wrongly
D) The antecedent almost never appears in the data

Answer: A. High confidence with a lift barely above one is the signature of a popular consequent. The antecedent is contributing almost nothing.

Q28. Apply the Isolation Forest idea to two records, one inside a dense cluster and one far from everything, and determine which gets the shorter average path length.

A) The record inside the dense cluster
B) The record far from everything <-- correct
C) The two get identical path lengths
D) The path length does not depend on position

Answer: B. Few random cuts are needed to isolate a point standing alone, and a short average path is exactly what the method reads as an anomaly.

Q29. Compare Apriori with FP-Growth on the number of passes each makes over the transaction database.

A) FP-Growth makes many passes, Apriori makes two
B) Both of the two make exactly two passes
C) Both of the two make one pass per itemset
D) Apriori makes many passes, FP-Growth makes two <-- correct

Answer: D. Apriori rescans the database once for every itemset size. FP-Growth compresses the data into a tree after two passes and then mines the tree.

Q30. Use a contamination setting of 0.01 on a dataset of 50,000 records to determine how many records will be flagged.

A) 50
B) 500 <-- correct
C) 5,000
D) 100

Answer: B. The setting is a proportion, so 0.01 x 50,000 = 500 records are flagged, whatever the underlying truth may be.

Q31. Solve the following: determine the lift when the rule support is 0.18, the antecedent support 0.30 and the consequent support 0.30.

A) 0.6
B) 1.0
C) 2.0 <-- correct
D) 0.09

Answer: C. Lift is 0.18 / (0.30 x 0.30) = 0.18 / 0.09 = 2.0, so the pair occurs twice as often as chance would predict.

Q32. Apply the Local Outlier Factor and determine what a reported value of 2.5 indicates about a record.

A) It sits in a region far sparser than its neighbours <-- correct
B) It sits in a region far denser than its neighbours
C) It is exactly as dense as its neighbours
D) It is a duplicate of one of its neighbours

Answer: A. A value well above 1 means the local density around the record is much lower than around the points it is compared with, which marks it as an outlier.

Q33. Examine a market basket result in which every high-confidence rule has the same popular staple as its consequent, and determine what has happened.

A) Confidence has been inflated by that item's base rate <-- correct
B) The minimum support was set far too low
C) The transactions were recorded incorrectly
D) The algorithm failed to prune its candidates

Answer: A. A very common consequent follows almost any antecedent. Filtering by lift instead of confidence removes these uninformative rules.

Q34. Use the level-wise structure of Apriori to determine which candidates it examines first.

A) The largest itemsets, before any smaller ones
B) The itemsets holding the most popular item
C) The itemsets appearing in the first transaction
D) The one-item sets, before any larger ones <-- correct

Answer: D. Apriori works upwards by size, using the frequent sets of one level to generate the candidates of the next, which is what makes the pruning rule usable.

Q35. Compute the support count corresponding to a minimum support of 0.04 on a database of 2,500 transactions.

A) 40
B) 100 <-- correct
C) 250
D) 625

Answer: B. The threshold in counts is 0.04 x 2,500 = 100 transactions.

Q36. Apply the anti-monotone property to determine the support of an itemset relative to the support of any of its subsets.

A) It is never smaller than that of any subset
B) It is always exactly equal to that of a subset
C) It is never greater than that of any subset <-- correct
D) It bears no relation to that of a subset
Answer: C. Every transaction containing the larger set must contain each subset, so adding an item can only keep the count the same or reduce it.

Q37. Compare anomaly detection with novelty detection on the data each one is trained upon.

A) Novelty detection trains on contaminated data, anomaly on clean
B) Both of the two train on data known to be clean
C) Anomaly detection trains on contaminated data, novelty on clean <-- correct
D) Both of the two train on data known to be contaminated
Answer: C. Novelty detection is given a reference set of normal records and asked to flag departures. Anomaly detection has to find the odd records inside the data it is given.

Q38. Examine the effect on an anomaly detector of lowering the threshold on its anomaly score.

A) Fewer records are flagged and false positives rise
B) More records are flagged and false positives rise <-- correct
C) More records are flagged and false positives fall
D) The number of flagged records does not change
Answer: B. A lower bar admits more records to the alert list, which catches more genuine anomalies but also sweeps in more ordinary ones.

Q39. Use the properties of Isolation Forest and Local Outlier Factor to select a method for scoring ten million records quickly.

A) Local Outlier Factor, which needs no neighbours
B) Local Outlier Factor, whose cost is always lower
C) Neither, since both require the full distance matrix
D) Isolation Forest, whose cost grows roughly linearly <-- correct
Answer: D. LOF must find the neighbours of every record, which is costly at this scale. Isolation Forest builds a small number of shallow random trees instead.

Q40. Demonstrate how the output of a market basket analysis is used inside a store.

A) Associated items are placed together or offered as a bundle <-- correct
B) Customers are ranked by their total spending
C) Products with low support are removed at once
D) Every transaction is checked for possible fraud
Answer: A. The point of the rules is action: shelf placement, bundle pricing, cross-sell prompts at the checkout and the layout of an online recommendation panel.

----- End of Sample Question Set -----